A reflexive counterexample to the outside-c_0 spaceability question

Solution packet for arXiv:2406.06859v2

12 August 2026

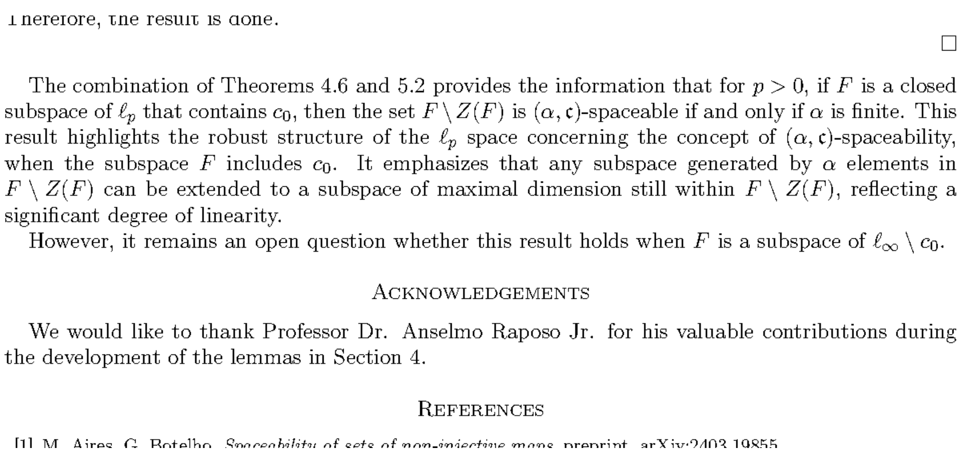
Status. Candidate full counterexample, likely valid, awaiting human review.

1 Source question and its interpretation

For a sequence space $F \subseteq \mathbb{K}^{\mathbb{N}}$, Alves and Ribeiro write [1]

$$Z(F) = \{x \in F : x \text{ has only finitely many zero coordinates}\}.$$

They prove, under a hypothesis involving c_0, that $F \setminus Z(F)$ is $(\alpha, \mathfrak{c})$-spaceable if and only if α is finite, and end with the following question on source PDF page 20.


Therefore, the result is done.

□

The combination of Theorems 4.6 and 5.2 provides the information that for $p > 0$, if F is a closed subspace of ℓ_p that contains c_0, then the set $F \setminus Z(F)$ is $(\alpha, \mathfrak{c})$-spaceable if and only if α is finite. This result highlights the robust structure of the ℓ_p space concerning the concept of $(\alpha, \mathfrak{c})$-spaceability, when the subspace F includes c_0. It emphasizes that any subspace generated by α elements in $F \setminus Z(F)$ can be extended to a subspace of maximal dimension still within $F \setminus Z(F)$, reflecting a significant degree of linearity.

However, it remains an open question whether this result holds when F is a subspace of $\ell_\infty \setminus c_0$.

Acknowledgements

We would like to thank Professor Dr. Anselmo Raposo Jr. for his valuable contributions during the development of the lemmas in Section 4.

References

[1] M. Aires, G. Botelho, *Spaceability of sets of non-injective maps*, preprint, arXiv:2403.19855

Literally, no vector subspace can be contained in $\ell_\infty \setminus c_0$, because every vector subspace contains $0 \in c_0$. The standard coherent reading is

$$F \cap c_0 = \{0\}, \qquad \text{equivalently} \qquad F \setminus \{0\} \subseteq \ell_\infty \setminus c_0. \tag{1}$$

One might instead read the question as merely requiring $c_0 \not\subseteq F$, or even as requiring that F contain no subspace isomorphic to c_0. The same counterexample satisfies all three interpretations.

Recall that a set A in a topological vector space is (α, β)-spaceable if it is α-lineable and every α-dimensional subspace $E \subseteq A \cup \{0\}$ is contained in a closed β-dimensional subspace $G \subseteq A \cup \{0\}$.

2 Proof intuition

Embed ℓ_2 isometrically into ℓ_∞ using a countable norming family of functionals. Instead of recording each functional once, repeat its value on an infinite coordinate block, and reserve one further infinite block on which every image is zero. A nonzero image cannot tend to zero because at least one nonzero functional value repeats infinitely often, so the range meets c_0 only at zero. Yet the reserved block gives every image infinitely many zero coordinates. Thus $Z(F)$ is empty and the exotic set $F \setminus Z(F)$ is simply the entire closed space F.

3 The counterexample

Theorem 1. *Over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, there exists a closed subspace $F \subseteq \ell_\infty$ such that:*

1. *F is linearly isometric to ℓ_2, hence is reflexive and contains no subspace isomorphic to c_0;*
2. *$F \cap c_0 = \{0\}$;*
3. *$Z(F) = \varnothing$.*

Consequently, for every cardinal $1 \le \alpha \le \mathfrak{c}$, the set $F \setminus Z(F)$ is $(\alpha, \mathfrak{c})$-spaceable. In particular the finite-α characterization proposed in the source question does not extend to subspaces outside c_0.

Proof. Let $H = \ell_2$ and choose a sequence $(\varphi_k)_{k \ge 1}$ in the unit sphere of H^* which is norming:

$$\sup_{k \ge 1} |\varphi_k(x)| = \|x\|_2 \qquad (x \in H). \tag{2}$$

For example, identify H^* with H and take the functionals induced by a countable dense subset of the unit sphere. Density and Cauchy–Schwarz give (2).

Partition $\mathbb{N}$ into pairwise disjoint infinite sets

$$\mathbb{N} = B \,\dot\cup\, A_1 \,\dot\cup\, A_2 \,\dot\cup \cdots .$$

One explicit choice is

$$B = \{2m - 1 : m \ge 1\}, \qquad A_k = \{2^k(2m-1) : m \ge 1\} \quad (k \ge 1).$$

Define $J : H \to \ell_\infty$ coordinatewise by

$$(Jx)_n = \begin{cases} 0, & n \in B, \\ \varphi_k(x), & n \in A_k. \end{cases} \tag{3}$$

This map is linear, and (2) gives

$$\|Jx\|_\infty = \sup_{k \ge 1} |\varphi_k(x)| = \|x\|_2.$$

Thus J is an isometry. Its range $F = J(H)$ is closed in ℓ_∞.

Suppose $Jx \in c_0$. For each fixed k, the coordinates of Jx are equal to the constant $\varphi_k(x)$ along the infinite, hence unbounded, set A_k. Convergence to zero forces $\varphi_k(x) = 0$ for every k. Equation (2) then gives $x = 0$. Therefore $F \cap c_0 = \{0\}$.

On the other hand, (3) makes every Jx vanish at every coordinate in the infinite set B. No member of F has only finitely many zero coordinates, so $Z(F) = \varnothing$. Hence

$$F \setminus Z(F) = F. \tag{4}$$

The Hamel dimension of H is $\mathfrak{c}$. For completeness, $|H| = \mathfrak{c}$ gives the upper bound, while the vectors

$$v_t = (t^n)_{n\geq 1} \in \ell_2 \qquad (0 < t < 1)$$

form a linearly independent family of cardinality $\mathfrak{c}$: every finite subfamily is independent by the Vandermonde determinant. Thus $\dim F = \mathfrak{c}$.

Now fix $1 \leq \alpha \leq \mathfrak{c}$ and any α-dimensional subspace $E \subseteq (F \setminus Z(F)) \cup \{0\} = F$. By (4), the space F itself is a closed $\mathfrak{c}$-dimensional subspace satisfying

$$E \subseteq F \subseteq (F \setminus Z(F)) \cup \{0\}.$$

This is exactly $(\alpha, \mathfrak{c})$-spaceability.

Finally, F is isometric to the Hilbert space H, so it is reflexive. Since c_0 is not reflexive and a Banach-space isomorphic embedding has closed range, F contains no subspace isomorphic to c_0. This also verifies the strongest interpretation discussed after (1). $\square$

4 Scope, verification, and novelty

The theorem supplies a full counterexample rather than only exploiting the literal defect in the phrase "subspace of $\ell_\infty \setminus c_0$." It satisfies the repaired condition (1), the weaker condition $c_0 \nsubseteq F$, and the stronger isomorphic-copy exclusion.

A bounded literature check through 12 August 2026 searched the current arXiv record and v2 full text, the exact title and arXiv id, the exact open-question wording, citation-oriented results, and close searches involving $F \cap c_0$, infinitely many zero coordinates, and $(\alpha, \mathfrak{c})$-spaceability. No later answer was found. This is a bounded web/arXiv novelty screen, not an exhaustive bibliographic guarantee.

Human-review recommendation. High priority. Check (2), closedness of the isometric range, the two block arguments, and the final direct use of the definition of $(\alpha, \mathfrak{c})$-spaceability.

References

[1] D. Alves and G. Ribeiro, *On Sequences with at Most a Finite Number of Zero Coordinates*, arXiv:2406.06859v2 [math.FA], 14 June 2024.